

written back on the disk, and the page is currently not in use.

- **ACTIVE** – the page is currently in use by some process.

Because memory is a limited resource, the kernel cannot store the dirty pages in RAM forever. As the kernel runs short of memory, the pages that have been dirtied, but are no longer used by a process, must be flushed back to the disk. This is handled by the *pdflush* kernel thread.

The default is a minimum of two threads. Use the following command to view the current status of *pdflush*:

```
[root@legacy ~]# sysctl -a | grep pdflush
```

```
vm.nr_pdflush_threads = 2
```

```
[root@legacy ~]
```

The idea behind *pdflush* is to avoid a sudden surprising load of heavy disk I/O. The goal is to write dirtied pages back to the disk at constant intervals. The default behaviour of *pdflush* is to write dirty pages to the disk as a kernel background task. The logic is simple: if the ratio of the number of pages that are dirty, to the total number of pages of the RAM available, reaches a certain percentage, *pdflush* writes the dirty pages synchronously.

These are some of the other parameters that you can use to tune the behaviour of *pdflush*:

1. Tuning as per the length/size of memory
 - *vm.dirty_background_ratio* – sets the percentage (of total memory) of dirty pages at which *pdflush* starts writing to disk.
 - *vm.dirty_ratio* – sets the percentage (of total memory) of dirty pages at which the process itself starts writing dirty data to disk.
2. Tuning according to time
 - *vm.dirty_writeback_centisecs* – sets the interval between *pdflush* wake-ups (in 1/100th of a second). Set zero to

disable *pdflush*.

- *vm.dirty_expire_centisecs* – sets the time when the data is old enough (in 1/100th of a second) to be written to disk by *pdflush*

Here is a sample default configuration for *pdflush* values on my system:

```
[root@legacy ~]# sysctl -a | grep vm.dirty
```

```
vm.dirty_background_ratio = 10
```

```
vm.dirty_background_bytes = 0
```

```
vm.dirty_ratio = 20
```

```
vm.dirty_bytes = 0
```

```
vm.dirty_writeback_centisecs = 500
```

```
vm.dirty_expire_centisecs = 3000
```

```
[root@legacy ~]
```

As you can see, *pdflush* will start writing the dirty pages to the disk as soon as 10 per cent of the total memory gets dirty. On the other hand, the process itself will start writing the data to disk as soon as 20 per cent of the total memory is dirty.

You can always reset these values by defining the new values in */etc/sysctl.conf* and then running *sysctl -p*.

Tuning swap

We all know what swap is and how it is important to us. We also know that swap is very slow compared to the physical memory (RAM). Naturally, when a system heavily relies on swap it basically means we need more RAM.

A common controversial question is how much swap space to dedicate to a system. The answer depends on several factors. However, generally, you can use the following guidelines:

- If the amount of RAM is less than 1 GB, the ideal swap space should be twice the size of RAM.
- If the amount of RAM is between 1-2 GB, the ideal swap space should be 1.5 times the size of


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